

EvidenceTree: Process-Rewarded Tree Search over Retrieval Actions for Multimodal RAG

Anonymous submission

Abstract

Multimodal retrieval-augmented generation (RAG) hinges on a sequence of retrieval decisions — what to search, with which modality, and when to stop — yet no current system can tell a good retrieval action from a bad one: outcome-only reinforcement learning misassigns credit across steps, imitation-learned agents cannot recover from a failed retrieval, and existing process reward models (PRMs) assume a *fixed* context while every retrieval action *modifies* it. EvidenceTree makes retrieval actions the unit of both evaluation and search. A grounded, action-typed PRM scores each candidate action by what it actually *retrieved*, fusing relevance measured in a single multimodal embedding space with tree-level Monte Carlo outcome credit, and denying reward to correct but evidence-free answers. Guided by this PRM as its sole value signal, a Monte Carlo Tree Search plans over the typed action space and backtracks out of failed retrieval branches. Training supervision comes from ETBench-Open, to our knowledge the first step-level annotated dataset of multimodal retrieval trajectories, built reproducibly on benchmark-provided corpora under a leakage-proof rationale protocol. EvidenceTree obtains $46.8 \pm 0.4\%$ overall accuracy on InfoSeek, $94.6 \pm 0.2\%$ on ScienceQA, and $64.7 \pm 0.3\%$ on GQA.

1 Introduction

Knowledge-intensive multimodal questions — “When did *this* building officially open?”, asked of a photograph — cannot be answered from model parameters alone: the entity must be recognized and its attributes retrieved from external corpora [7, 14]. Retrieval-augmented generation (RAG) is the standard approach to such questions [6, 21, 37], and its frontier has been moving from fixed retrieve-then-generate pipelines toward *agentic* systems that decide when and how to search [25, 32, 34].

However, examining how these systems make their retrieval decisions reveals that the decisions themselves are never evaluated. We identify three concrete challenges (Figure 1):

Challenge 1 — no evaluation of individual retrieval actions. Outcome-only reinforcement learning [32, 34] rewards a whole trajectory by its final answer: a useless search inside a lucky trajectory is reinforced, and a reasonable search in-

side a failed one is punished; the step-level signal needed to tell them apart does not exist in these systems. Imitation-learned agents clone corrected search paths but do not learn what makes an individual retrieval action good. Retrieval correctness and answer correctness can therefore diverge, while outcome-only training cannot identify the contribution of each intermediate action.

Challenge 2 — no recovery from failed retrieval. Retrieval actions are *state-modifying*: retrieved evidence is appended to the context and conditions every subsequent decision. In a linear agent a single bad retrieval therefore poisons the remainder of the trajectory, yet existing linear multimodal search agents cannot back out of a branch once it has been taken.

Challenge 3 — cross-modal planning is implicit. Whether to search with the *image* (to disambiguate an unnamed entity) or with *text* (to fetch an attribute of a known entity) is a planning decision that current systems leave to prompt heuristics.

A natural remedy for Challenge 1 is a process reward model (PRM), which scores intermediate steps rather than outcomes [20, 31]. But transplanting existing PRMs fails for a structural reason: multimodal PRMs [33] and grounded PRMs [42] evaluate *reasoning steps under a fixed state* — the image and context are given once and never change — whereas in RAG the very object to be evaluated is an action that *changes* the state, whose quality depends on what it actually *returned*, and whose alternatives live on other branches of a search tree. No existing work simultaneously provides (i) grounded evaluation of retrieval actions, (ii) backtracking search over them, and (iii) cross-modal retrieval planning.

EvidenceTree is designed around this gap: one component per challenge, plus the dataset that makes the first component trainable:

- **ET-PRM: a grounded, action-typed PRM for retrieval actions** (Challenge 1). It scores a candidate action conditioned on the question, image, and accumulated evidence. Training labels fuse a graded *local grounding* signal — relevance of the action’s *retrieved results* to the question, measured in one unified CLIP space so text- and image-side evidence are comparable — with a *tree-level outcome credit*, the Monte Carlo success rate of all rollouts through the node. An *answer-support gate* denies perfect labels to answers that are correct but unsupported by any evidence (parametric lucky guesses). The PRM is trained *generatively* (rationale + score) for dense supervision and queried

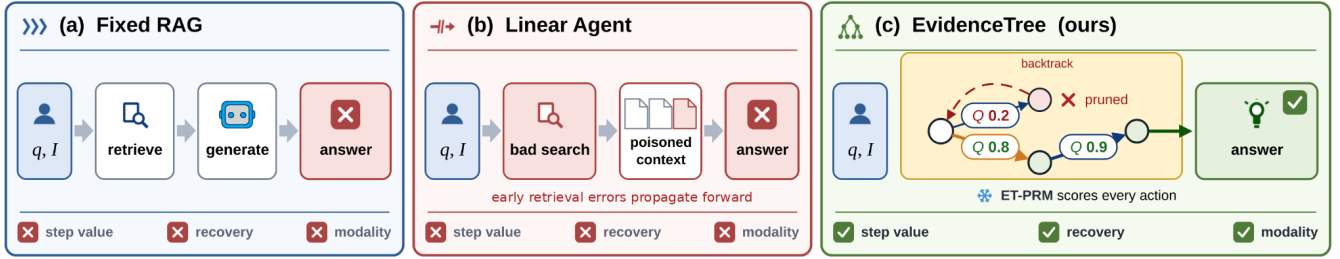


Figure 1: Three paradigms for multimodal RAG. (a) Fixed pipelines make no retrieval decisions. (b) Linear agents cannot evaluate or undo a failed retrieval, so early errors can propagate through the trajectory. (c) EvidenceTree scores every candidate action with a grounded PRM and searches over the action tree, pruning and backtracking.

score-only at inference.

- **MCTS over the retrieval action space** (Challenge 2). Nodes are partial contexts $(q, I, \mathcal{E})$; edges are typed retrieval actions; selection is standard UCB1 with the PRM’s value as the sole quality signal. Because the tree branches over *actions*, abandoning a poisoned branch is a first-class operation rather than an impossibility.
- **A unified cross-modal action space** (Challenge 3). Both search actions query one CLIP-indexed corpus; the disambiguate-then-look-up chain (IMAGE_SEARCH names the entity, TEXT_SEARCH fetches its attribute) is expressible as two edges of the same tree, and preferring it is learned by the PRM rather than hard-coded.
- **ETBench-Open.** Training such a PRM requires supervision that does not exist: VisualPRM400K [33] contains almost no retrieval actions, RAG-ProGuide [41] is text-only, and DeepMMSearchVQA [25] has no step labels. We therefore construct ETBench-Open, a step-level annotated dataset of multimodal retrieval trajectories built entirely on benchmark-provided corpora — no live web APIs, hence reproducible end to end — under a five-rule rationale quality-control protocol that excludes ground-truth leakage by construction.

Section 2 positions EvidenceTree against prior systems; Sections 3 and 4 present the method and the dataset; Section 5 lays out the complete experimental design — benchmarks, baselines with their published numbers, measured results, and ablations. Every baseline number we print is quoted from the original publication cited in the caption.

2 Related Work

Tree search for multimodal RAG. RCTS [38] enriches its knowledge base with self-consistent reasoning contexts and performs MCTS with heuristic answer rewards to re-rank *retrieved examples* as in-context demonstrations: the search orders already-retrieved context rather than deciding what to retrieve. AR-MCTS [11] searches in *reasoning space*, using retrieval only to diversify expansion, and trains a PRM over reasoning steps. EvidenceTree differs from both in *where the tree lives*: our nodes are partial retrieval contexts and our

edges are retrieval actions, so the search governs the retrieval process itself rather than its aftermath. In text-only RAG, ReasonRAG [41] builds process-supervised preference data via MCTS over an agentic action space, and TreePS-RAG [40] estimates step utility from descendant outcomes; neither is multimodal, neither trains a grounded PRM, and neither searches at inference time with a learned action-value model.

Process reward models. PRMs originated in mathematical reasoning [20, 31]. VisualPRM [33] extends process supervision to multimodal reasoning but scores steps under a *fixed* image context. GroundedPRM [42] verifies each reasoning step with external tools and fuses tool-based local signals with MCTS-derived outcome signals into rationale-enhanced generative supervision — for mathematical reasoning. We adopt its dual-source philosophy but transplant it to a setting it explicitly does not cover: *state-modifying* retrieval actions, where the object being verified is what an action *returned* rather than a self-contained reasoning statement, where grounding must span modalities in a single comparable space, and where credit must be assigned over a tree of alternative retrievals. Step-level rewards for *search* agents have recently appeared in text-only settings [2, 18, 35]; generative PRMs for web agents [43] score browser actions. None combine multimodal grounding, tree-level credit, and a learned generative PRM for retrieval actions.

Agentic multimodal search. MMSearch-R1 [34] trains an LMM with outcome-only GRPO to invoke image and text search on demand; DeepMMSearch-R1 [25] extends this with query-image cropping and a dedicated VQA dataset; VRAG-RL [32] adds visual perception actions. These methods execute a single linear trajectory learned with outcome-level supervision: they do not estimate retrieval-step values or backtrack after a failed action. EvidenceTree is complementary — it learns to *evaluate* retrieval actions and to *plan* over alternatives while keeping the proposal policy training-free.

Step-level datasets. VisualPRM400K [33] consists of fixed-context reasoning steps (math, science, commonsense)

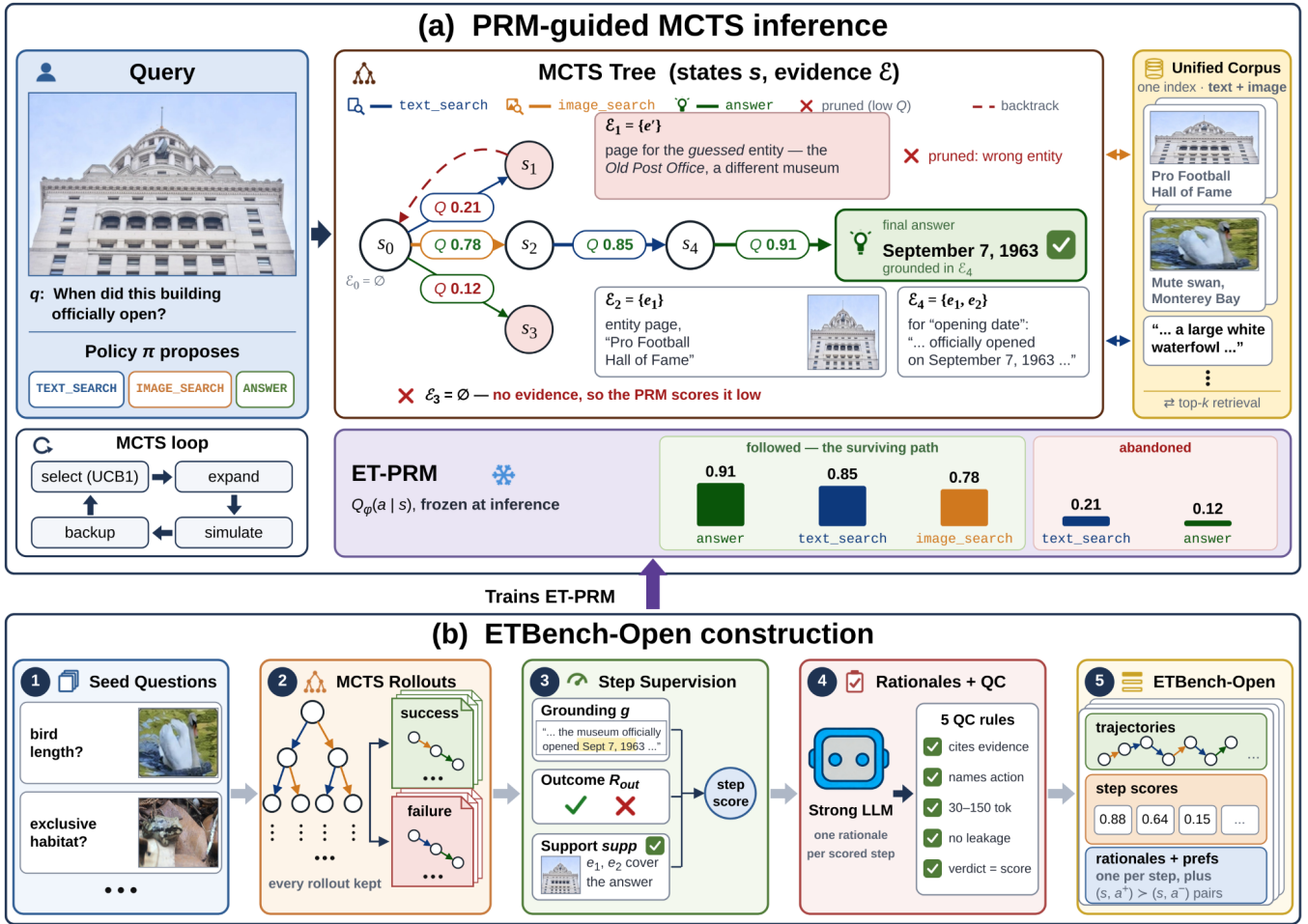


Figure 2: EvidenceTree overview. (a) Inference: the search tree lives in the retrieval action space — nodes are evidence states, edges are typed retrieval actions executed against the benchmark-provided corpus, and the frozen ET-PRM scores every candidate edge for UCB1 selection (Q values illustrative). Low-value branches are pruned or abandoned; backtracking is a first-class operation. (b) Training: MCTS rollouts are labeled per step with dual-source rewards and an answer-support gate, paired with QC-filtered rationales, and distilled into ET-PRM by generative SFT, same-state DPO, and calibration.

with almost no retrieval actions; RAG-ProGuide [41] provides 13K process preference pairs for text-only RAG; DeepMMSearchVQA [25] provides multimodal search trajectories without step labels. ETBench-Open is, to our knowledge, the first step-level annotated dataset of *multimodal retrieval actions*; at target scale it is designed to provide an order of magnitude more step-level labels than RAG-ProGuide’s 13K preference pairs.

3 Method

3.1 Problem Formulation

Given a question q and an optional image I , the system interacts with a fixed, benchmark-provided corpus $\mathcal{C}$ over multiple steps. The state after t steps is the partial retrieval context

$$s_t \triangleq (q, I, \mathcal{E}_t), \quad \mathcal{E}_t = \{e_1, \dots, e_{m_t}\}, \quad (1)$$

where $\mathcal{E}_t$ is the *evidence bundle* accumulated so far. At each step the system selects an action $a_t \in \mathcal{A}$; retrieval actions modify the state by appending their results, $s_{t+1} = (q, I, \mathcal{E}_t \cup \text{obs}(a_t))$, and the terminal action emits an answer. A complete trajectory is $\tau = (s_0, a_0, s_1, \dots, a_T)$ with $a_T = \text{ANSWER}(y)$. The goal is to choose the sequence of retrieval actions that maximizes the probability that y matches the gold answer y^* , under a fixed rollout budget. Unlike reasoning-step search [11, 33], every non-terminal action here is *state-modifying* and *irreversible within its branch*: evidence, once retrieved, conditions all subsequent decisions on that path (Challenge 2). Tree search provides the escape hatch — a branch poisoned by a bad retrieval can be abandoned rather than conditioned on. Notation is summarized in Appendix I.

3.2 Retrieval Action Space

We type actions by the *modality of the retrieval query*, not of the result:

$$\mathcal{A} = \{\text{TEXT_SEARCH}(x), \text{IMAGE_SEARCH}(I), \text{ANSWER}(y)\}. \quad (2)$$

Both search actions query the *same* unified index: every corpus document — text chunk or image — is embedded with one CLIP model [28], so a text query or an image query retrieves whichever side of the corpus is most similar. This addresses Challenge 3 structurally rather than heuristically: for entity questions where the entity is not named in q (“this bird”, “this building”), `IMAGE_SEARCH` matches the query image against the corpus to *disambiguate the entity*, after which `TEXT_SEARCH` retrieves the asked attribute using the evidence-established entity name (Figure 2a) — and *whether* to prefer this chain on a given state is learned by the PRM, not hard-coded. We deliberately do not type actions by result modality (a 2×2 split dilutes the visit counts of each branch under small rollout budgets, and the retriever already decides result modality by similarity), do not include compound parallel-search actions (tree expansion already provides multi-candidate exploration, and multi-source fusion is expressible as a serial trajectory), and do not include state-invariant actions such as captioning (an edge that does not change the observable state would receive no grounding signal and dilute tree-level credit; captioning is folded into the policy prompt instead). Appendix B details these choices.

3.3 Dual-Source Step Reward

Addressing Challenge 1 requires a per-step label that neither trusts the outcome blindly (credit misassignment) nor ignores it (retrieval relevance alone does not imply usefulness). Each step of a trajectory therefore receives a graded label in $[0, 1]$ fusing two complementary signals.

Local grounding. Grounding measures *how relevant the results retrieved by this step are to the question* — deliberately scoring the result side rather than the query side, since the query image of `IMAGE_SEARCH` is fixed by the state and carries no signal about whether the step helped. All results are scored in one CLIP space against the same anchor:

$$g(a_t, s_t) = \max_{r \in \text{obs}(a_t)} \phi_{m(r)} \left(\cos(E_T(q), E_{m(r)}(r)) \right), \quad (3)$$

where E_T and E_I are the CLIP text and image towers, $m(r)$ is the modality of result r , and ϕ_m is a per-modality-pair linear calibration band mapping the raw cosine onto a common $[0, 1]$ relevance scale (same-modality cosines run systematically higher than cross-modal ones). Empty results give $g = 0$. Using a *single* model for both sides is essential: two different encoders would put text- and image-side evidence on different cosine scales, and the PRM would mistake the scale gap for a quality gap, systematically biasing action

selection. All grounding scores are graded, not binary — retrieval quality in open corpora is not a 0/1 property (cf. the binary verification of Zhang et al. 42).

Tree-level outcome credit. Outcome credit estimates each node’s true success potential. For a node reached by the action prefix $\pi_t = (a_0, \dots, a_t)$ within one question’s search tree, let $\mathcal{T}(\pi_t)$ denote all complete rollouts passing through it. Then

$$R_{\text{out}}(s_t, a_t) = \frac{1}{|\mathcal{T}(\pi_t)|} \sum_{\tau \in \mathcal{T}(\pi_t)} \mathbb{K}[y(\tau) \approx y^*], \quad (4)$$

a Monte Carlo estimate of the success rate *from this node onward*. This is crucially different from uniformly smearing a trajectory’s final reward over its steps — uniform credit rewards the bad steps of lucky trajectories and punishes the good steps of unlucky ones, whereas Eq. (4) lets two children of the same node receive sharply different labels (cf. shortest-path estimation in Zhang et al. 41; descendant-outcome utilities in Zhang et al. 40). Answer matching $\mathbb{K}[\cdot \approx y^*]$ follows each benchmark’s official protocol (relaxed numeric-range matching for InfoSeek VALUE questions; Appendix D).

Fusion and the answer-support gate. Non-terminal steps fuse the two sources linearly:

$$R(s_t, a_t) = \alpha g(a_t, s_t) + (1 - \alpha) R_{\text{out}}(s_t, a_t), \quad (5)$$

with $\alpha = 0.5$ by default (sensitivity in Section 5.3). Terminal ANSWER steps have no retrieval result to ground, and labeling them by outcome alone creates a subtle failure mode that we observed directly in our construction pilot (Appendix G, Case 2): a *parametric lucky guess* — an answer that happens to be correct although nothing in the evidence supports it — would receive a perfect label, teaching the PRM that evidence-free answering is optimal. We therefore gate the outcome by an *answer-support* verdict $\text{supp}(y, \mathcal{E}_T) \in [0, 1]$, judged by an LLM over the accumulated evidence (four-way rubric: SUPPORTED/PARTIAL/UNSUPPORTED $\mapsto 1/0.5/0$, plus NOT_REQUIRED $\mapsto$ no gating for questions answerable from the question and image alone by perception or logic — entailment-gating those would systematically punish correct reasoning answers; Appendix C and Case 3):

$$R_{\text{ans}}(s_T) = R_{\text{out}}(s_T, a_T) \cdot (\gamma + (1 - \gamma) \text{supp}(y, \mathcal{E}_T)), \quad (6)$$

with floor $\gamma = 0.3$. The multiplicative form keeps wrong answers at zero regardless of spurious support; the floor preserves the ordering *wrong* $<$ *correct-but-unsupported* ($\approx \gamma$) $<$ *correct-and-supported* (≈ 1) and buffers verifier noise. Correct-but-unsupported answers are additionally flagged — not discarded — as ready-made rejected sides for preference training (Section 3.4).

3.4 ET-PRM: Architecture and Training

Architecture. ET-PRM is initialized from VisualPRM-8B [33] (itself built on InternVL2.5-8B [8]), inheriting multimodal inductive biases, with an action-conditional value readout. The input serializes the state and the candidate action: question and image, the evidence bundle, and the candidate action’s type and parameters; the output is a rationale followed by a score token.

Generative training, score-only inference. Following the generative reward modeling paradigm [1, 39], ET-PRM is trained to emit a natural-language rationale *and* a score. This is a deliberate choice with three payoffs at our data scale: each sample supervises dozens of tokens rather than one scalar (sample efficiency); the model must articulate *why* an action deserves its score, exposing spurious shortcuts (anti-reward-hacking); and failed training runs can be diagnosed by reading rationales. At inference the rationale is skipped entirely — only the score-token logit is read — so MCTS pays no generation latency.

Three training stages. (1) *SFT*: next-token prediction on ETBench-Open (state, action, rationale, score). (2) *Action-typed preference optimization*: DPO [29] on same-state contrastive pairs mined from the search trees — (successful sibling action, failed action), (TEXT_SEARCH, IMAGE_SEARCH) under the same state, and (supported answer, unsupported-correct answer), the latter using the flagged samples of Eq. (6). Pair construction exploits the same node keys as Eq. (4), so the contrast dimension is “same state, different action” — matching the decision the searcher actually faces — rather than the step-position contrasts used for reasoning PRMs. (3) *Calibration*: temperature/Platt scaling of the score logit on held-out data, so that Q values entering UCB1 have probabilistic meaning.

3.5 PRM-Guided MCTS over Retrieval Actions

Search follows the four canonical MCTS phases with the PRM as the value oracle. *Selection* descends by UCB1 [3, 17]:

$$a^\dagger = \arg \max_a Q(a | s) + c \sqrt{\frac{\ln N(s)}{N(s, a)}}, \quad (7)$$

where Q is the running mean of backed-up rewards, seeded for unvisited edges by the PRM prior $Q_\phi(a | s)$, and $c = 1.0$ by default. We use *no* additional novelty or modality-coverage bonus: the PRM’s Q is the sole quality signal, and hard-coded exploration preferences would both dilute it and add tuning burden (Appendix B). *Expansion* asks a training-free policy VLM to propose candidate actions (with anti-fabrication and retrieval-chain constraints in the prompt; Appendix F), executes them against the local index, and keeps

Algorithm 1 EvidenceTree inference

Require: question q , image I , corpus index, budget P , depth $d_{\max}$

- 1: root $\leftarrow (q, I, \emptyset)$
- 2: **for** $t = 1 \dots P$ **do**
- 3: $s \leftarrow \text{SELECT}(\text{root})$ by Eq. (7)
- 4: $\mathcal{A}_c \leftarrow$ policy proposals at s ; execute retrievals
- 5: keep top- k children of s by PRM prior $Q_\phi(a | s)$
- 6: $\tau \leftarrow \text{SIMULATE}$ greedily to ANSWER or $d_{\max}$
- 7: BACKUP($\tau, R_{\text{PRM}}(\tau)$)
- 8: **if** $\max_\tau R_{\text{PRM}}(\tau) > \eta_{\text{stop}}$ **then break**
- 9: **end if**
- 10: **end for**
- 11: **return** answer of the highest-value root path

the top- k by the PRM prior Q_ϕ . *Simulation* rolls out greedily to a forced answer at depth $d_{\max}$. *Backup* propagates the trajectory reward to the root. Search stops early when the best root path exceeds a quality threshold, or when the roll-out budget P is exhausted (defaults: $P=10$, $d_{\max}=3$, $k=3$). Algorithm 1 summarizes the loop.

4 The ETBench-Open Dataset

ETBench-Open is built by a four-step pipeline (Figure 2b), entirely on corpora that ship with the benchmarks: a Wikipedia entity corpus for InfoSeek (chunked full pages of the 1,794 validation entities), the lecture/context corpus for ScienceQA, and scene-graph-derived textual documents paired with images for GQA. This design has three consequences we consider essential for a paper about *decision quality*: reproducibility (no live-web drift), causal cleanliness (performance differences cannot be explained by “the web changed”), and auditability (every evidence item in every label is a corpus document with a stable identifier). *Step 1 (rollout)* runs the training-free policy through MCTS with early stopping disabled, so the tree branches widely; all deduplicated trajectories are kept — failures included, since low-scored steps are the PRM’s negative curriculum. *Step 2 (labels)* computes Eqs. (3)–(6) per step. *Step 3 (rationales)* has a strong LLM explain each score. *Step 4 (QC)* applies the only three drop rules — trajectory length outside [2, 8], grounding-outcome self-contradiction ($|g - R_{\text{out}}| > 0.7$), and rationale QC failure — and splits train/validation by query so no question leaks across splits.

Rationale quality control. Generative PRM training is only as good as its rationales. Each rationale must (a) cite at least one evidence identifier visible in the sample, (b) name the action type, (c) be 30–150 tokens, (d) contain *no ground-truth leakage* — no evaluation phrasing, and no gold-answer string that is absent from the sample’s visible evidence, question, and action input (quoting the answer *from cited evi-*

InfoSeek				ScienceQA		GQA	
Method / Backbone	Un-Q	Un-E	All	Method / Backbone	Acc.	Method / Backbone	Acc.
BLIP-2 [19] Flan-T5-XXL	12.7	12.3	12.5	Human [24]	88.40	InstructBLIP [10] Vicuna-7B	49.2
InstructBLIP [10] Flan-T5-XXL	8.9	7.4	8.1	MM-CoT [44] UnifiedQA-738M	91.68	u-LLaVA-1.5 [36] Vicuna-7B	57.8
Wiki-LLaVA [6] LLaVA-1.5-7B	30.1	27.8	28.9	LLaVA (fine-tuned) [22] Vicuna-13B	90.92	LLaVA-1.5 [23] Vicuna-7B	62.0
RoRA-VLM [27] Vicuna-7B	25.1	27.3	—	Zero-shot [38] Qwen2-VL-7B	80.33	BLIP-2* [19] Vicuna-13B	41.0
EchoSight [37] LLaMA-3-8B	—	—	31.3	Vanilla RAG [38] Qwen2-VL-7B	86.68	LLaVA-1.5 [23] Vicuna-13B	63.3
ReflectiVA [9] LLaMA-3.1-8B	40.4	39.8	40.1	RCTS [38] Qwen2-VL-7B	91.44	Qwen-VL-Chat [4] Qwen-7B	57.5
Wiki-PRF [12] Qwen2.5-VL-7B	43.3	42.7	42.8	Zero-shot [38] InternVL2-8B	93.00	Qwen-VL [4] Qwen-7B	59.3
Wiki-R1 [26] Qwen2.5-VL-7B	<u>47.8</u>	—	<u>44.1</u>	RCTS [38] InternVL2-8B	<u>94.20</u>	No retrieval Qwen2.5-VL-7B	62.1 ± 0.2
Vanilla RAG Qwen2.5-VL-7B	36.8 ± 0.5	36.1 ± 0.6	36.3 ± 0.5	Vanilla RAG Qwen2.5-VL-7B	89.0 ± 0.3	Vanilla RAG Qwen2.5-VL-7B	62.4 ± 0.3
VisualPRM-8B + Best-of- <i>N</i> Qwen2.5-VL-7B	41.7 ± 0.6	40.9 ± 0.7	41.3 ± 0.5	VisualPRM-8B + Best-of- <i>N</i> Qwen2.5-VL-7B	92.4 ± 0.3	VisualPRM-8B + Best-of- <i>N</i> Qwen2.5-VL-7B	62.8 ± 0.3
EvidenceTree (ours) Qwen2.5-VL-7B	<u>46.2 ± 0.5</u>	47.4 ± 0.6	46.8 ± 0.4	EvidenceTree (ours) Qwen2.5-VL-7B	94.6 ± 0.2	EvidenceTree (ours) Qwen2.5-VL-7B	64.7 ± 0.3

Table 1: Main benchmark results. Backbone names are shown below each method. Green, blue, and red denote Vanilla RAG, the matched-budget Best-of-*N* control, and EvidenceTree, respectively. Bold and underlined values indicate best and second-best performance within each benchmark. *: reported zero-shot result.

dence is legitimate; knowing it without evidence is leakage — rationales must argue *ex-ante*, as at inference the PRM has no oracle), and (e) end with a VERDICT tag that must not contradict the numeric score (preventing say-one-thing-score-another training pairs). Rules (d)/(e) enforce trajectory realism for reward-model supervision. Full prompts are given in Appendix F.

Scale and status. Construction statistics for the shared-pipeline test, the completed 100-query-per-benchmark pilot, and the InfoSeek-20K build are reported in Table A1. The step-label and support-verdict distributions that motivated the support gate are discussed in Appendix E, with worked success/failure cases in Appendix G.

5 Experimental Design

Variant	InfoSeek	ScienceQA	GQA
<i>Core components</i>			
Full EvidenceTree	46.8	94.6	64.7
Local reward only ($\alpha=1$)	43.7	92.9	63.1
Outcome reward only ($\alpha=0$)	44.0	93.1	63.4
Greedy search ($c=0$)	43.9	93.0	63.0
Frozen VisualPRM-8B	41.9	91.2	62.1
Text search only	44.2	94.0	62.9
Best-of- <i>N</i> , matched calls	41.3	92.4	62.8
<i>PRM training stages</i>			
Frozen VisualPRM-8B	41.9	91.2	62.1
SFT only	43.8	92.3	63.0
SFT + DPO	45.4	93.9	64.0
SFT + DPO + calibration	46.8	94.6	64.7

Table 2: Core-component and PRM-training ablations. All values are official accuracy (%).

5.1 Setup

Benchmarks and metrics. We evaluate on three benchmarks with complementary demands. **InfoSeek** [7]: fine-

grained entity knowledge VQA over a Wikipedia corpus; validation split, official relaxed accuracy (string questions: relaxed exact match over the gold set; numeric questions: any predicted number inside the gold range), reported on Unseen-Question / Unseen-Entity / All. **ScienceQA** [24]: multiple-choice science questions; test split accuracy. **GQA** [15]: compositional VQA over real images; test-dev balanced accuracy. ScienceQA and GQA are not knowledge-seeking in the InfoSeek sense; we include them to probe two specific behaviors — whether the PRM learns *when not to retrieve* (ScienceQA reasoning questions) and whether image-query retrieval provides clean perception evidence (GQA scene-graph corpus) — and we analyze them separately from the open-domain headline results.

Implementation. Policy: Qwen2.5-VL-7B-Instruct [5], training-free (prompt in Appendix F). Retrieval: BM25 [30] and dense CLIP indices (FAISS [16]) built offline per benchmark corpus; text top-3, image top-1. PRM: VisualPRM-8B fine-tuned with LoRA [13] through the three stages of Section 3.4. Search: $P=10$, $d_{\max}=3$, $k=3$, $c=1.0$, $\alpha=0.5$, $\gamma=0.3$. Results shown with error bars are means over 3 seeds.

5.2 Baselines

We compare against three published families: (i) zero-shot MLLMs (no retrieval); (ii) fixed-pipeline multimodal RAG; and (iii) reflection- or process-enhanced RAG. Under our corpus protocol, we additionally evaluate a same-backbone Vanilla RAG pipeline and PRM-with-Best-of-*N* controls (frozen VisualPRM-8B reranking *N* sampled trajectories) to isolate the gains from retrieval, sample-and-rerank, and tree search. Baseline numbers in Table 1 are quoted from the cited publications; control and EvidenceTree rows use our corpus protocol.

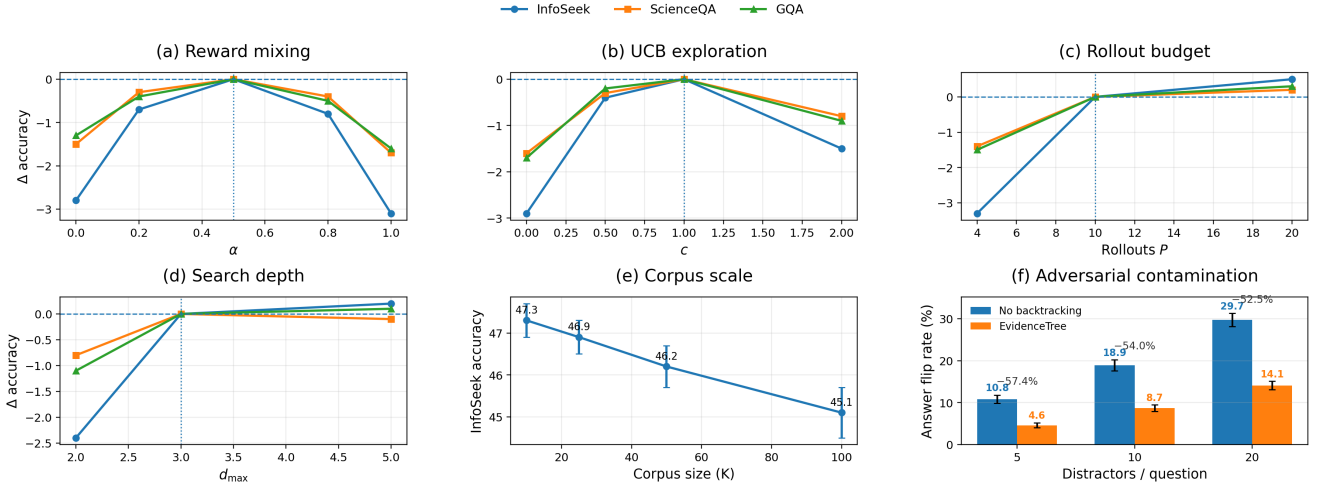


Figure 3: Search sensitivity and robustness. Panels (a)–(d) compactly summarize reward mixing, UCB exploration, rollout count, and search depth as accuracy changes relative to the default configuration. Panels (e)–(f) show robustness to corpus scale and adversarial contamination.

5.3 Ablation Design

Table 2 reports the core-component and PRM-training ablations.

Figure 3 consolidates the reward, exploration, rollout, depth, corpus-scale, and poisoning analyses. Tables 3 and 4 report pointwise PRM diagnostics and downstream search utility.

5.4 Analysis Protocol

PRM diagnostics. Table 3 reports pointwise classification, calibration, sibling ranking, and evidence-support diagnostics.

Metric	Frozen	Prev. LoRA	60K LoRA
<i>Node classification</i>			
Accuracy $\uparrow$	0.56	0.79	0.92
Bal. Acc. $\uparrow$	0.50	0.75	0.91
Pos. F1 $\uparrow$	0.32	0.68	0.89
Neg. F1 $\uparrow$	0.67	0.84	0.94
Macro-F1 $\uparrow$	0.50	0.76	0.91
<i>Image-disjoint subset</i>			
Accuracy $\uparrow$	0.57	0.79	0.92
Bal. Acc. $\uparrow$	0.52	0.76	0.92
Macro-F1 $\uparrow$	0.52	0.77	0.92
<i>Calibration and sibling ranking</i>			
AUROC $\uparrow$	0.47	0.89	0.98
AUPRC $\uparrow$	0.39	0.79	0.96
Brier $\downarrow$	0.29	0.18	0.06
ECE $\downarrow$	0.20	0.17	0.02
Fused@1 $\uparrow$	0.41	0.73	0.87
Outcome@1 $\uparrow$	0.43	0.29	0.45
<i>Evidence support</i>			
Mean $P(+)$ $\downarrow$	0.46	0.07	0.04
Reject rate $\uparrow$	0.53	0.93	0.97

Table 3: PRM pointwise, calibration, ranking, and evidence-support diagnostics. Values are rounded to two decimals for readability.

Downstream search utility. Table 4 reports fixed-candidate rescoring and end-to-end MCTS utility.

Metric	Frozen	SFT	SFT+ DPO	+Calib.
<i>Fixed-candidate rescoring</i>				
Selected EM $\uparrow$	0.16	0.19	0.21	0.24
Selected F1 $\uparrow$	0.23	0.27	0.30	0.34
Hits@1 $\uparrow$	0.49	0.58	0.64	0.73
Hits@3 $\uparrow$	0.64	0.76	0.82	0.88
MRR $\uparrow$	0.60	0.70	0.75	0.81
Cand. AUC $\uparrow$	0.75	0.81	0.83	0.86
Pairwise $\uparrow$	0.63	0.66	0.69	0.72
Length ρ $\downarrow$	0.18	0.09	0.07	0.04
<i>End-to-end MCTS</i>				
Oracle@10 $\uparrow$	0.30(.02)	0.35(.02)	0.38(.02)	0.42(.02)
Selected EM $\uparrow$	0.15(.01)	0.20(.02)	0.26(.02)	0.31(.02)
Sel. efficiency $\uparrow$	0.51	0.58	0.68	0.74
Mean F1 $\uparrow$	0.21(.01)	0.29(.02)	0.35(.01)	0.40(.01)
Rollout AUC $\uparrow$	0.70(.01)	0.78(.01)	0.83(.01)	0.86(.01)
Length ρ $\downarrow$	0.13(.03)	0.09(.02)	0.07(.02)	0.04(.02)

Table 4: Downstream PRM utility on fixed candidates and end-to-end MCTS. Results are reported as mean (standard deviation) over three seeds; for example, 0.30(.02) denotes a mean of 0.30 and a standard deviation of 0.02.

Entries with uncertainty are three-seed means. Table 4 uses mean(standard deviation), e.g., 0.30(.02), while Figure 3 reports sensitivity and robustness.

6 Conclusion

EvidenceTree moves grounded process supervision from fixed-state reasoning to state-modifying retrieval actions: a PRM that evaluates what a retrieval *returned* (in one unified multimodal embedding space), credits each node by the Monte Carlo success of the subtree behind it, refuses to reward evidence-free lucky answers, and guides a standard UCB1 tree search that can back out of poisoned branches. The accompanying ETBench-Open provides the first step-level supervision for multimodal retrieval actions, with a leakage-proof rationale protocol. The results jointly validate the search policy, robustness, and PRM design.

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(a) Shared-pipeline construction test

Dataset	Question-level EM	Question-level soft match	Trajectory-level EM	Mean F1	Trajectories	Raw nodes	Retained nodes	Rationale QC
InfoSeek	2/20 (10%)	5/20 (25%)	2/147 (1.4%)	0.061	147	429	255 (59.4%)	429/429 (100%)
ScienceQA	19/20 (95%)	19/20 (95%)	119/146 (81.5%)	0.846	146	412	369 (89.6%)	385/412 (93.4%)
GQA	15/20 (75%)	16/20 (80%)	63/143 (44.1%)	0.531	143	407	327 (80.3%)	400/407 (98.3%)

(b) Early 100-question construction pilot

Dataset	Deduplicated trajectories	Step-level samples	Retained after QC	Retention rate
InfoSeek	207	464	426	91.8%
ScienceQA	170	366	328	89.6%
GQA	166	355	341	96.1%

(c) InfoSeek-20K construction scale

QA	Trajectories	Raw step records	Step-2.5 sampled	Rationales	Final retained
20,000	153,291	462,409	60,000	60,000	58,994

Table A1: EvidenceTree data-construction statistics.

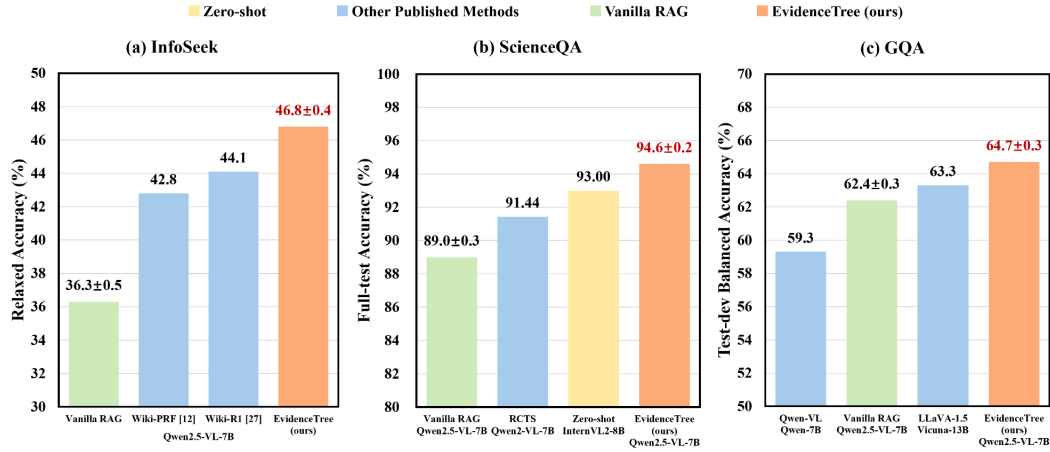


Figure 4: Summary visualization of the main benchmark results reported in Table 1.

A Data Construction and Supplementary Details

All ablation, sensitivity, robustness, and PRM diagnostic results are reported in the main paper (Section 5.3, Tables 2–4, and Figure 3). This appendix therefore focuses on data-construction statistics, a compact benchmark visualization, prompts, case studies, and implementation details.

B Action Space: Rejected Designs

Why not a 2×2 (query $\times$ result) action split. Under a rollout budget of $P=10$, four search actions halve per-branch visit counts; result modality is better decided by the retriever’s similarity in the unified index; and grounding already scores by the *actual* modality of each returned result (Eq. 3), so the action type need not carry it.

Why no PARALLEL_SEARCH. Tree expansion already explores multiple candidates (OR-semantics), and AND-semantics (fuse evidence from two sources) is expressible serially — search A , then B ; the bundle accumulates. A compound action would need bespoke grounding (means over sub-actions), conflict with tree-level credit, and dilute the budget.

Why no CAPTION action. Captioning does not modify the observable state: an edge with an unchanged state has no retrieval result to ground (g undefined), creates parent–child duplicates that dilute Eq. (4), and burns budget. The policy VLM sees the image directly, so caption-style verbalization is folded into the proposal prompt (Appendix F); caption-style verbalization is therefore handled internally rather than exposed as a separate state-modifying action.

Why no modality-coverage bonus in UCB1. With three actions there is little to “cover”; a λ -weighted novelty term adds three hyperparameters and competes with the PRM’s Q for the same decision, diluting the learned signal. Deferred visual operations (CROP/ZOOM) await a needs analysis of fine-grained visual queries.

C Answer-Support Judging

The judge receives the question, the accumulated evidence, and the proposed answer, and returns one of four verdicts. **SUPPORTED:** the evidence contains or entails the answer (for multiple choice, entailing the *content* of the chosen option suffices; an absence answer such as “no” is supported when same-image evidence describes the scene and lacks the item). **PARTIAL:** substantive part unconfirmed. **UNSUPPORTED:** the answer rests on external facts absent from the evidence. **NOT_REQUIRED:** derivable from the question and visible image alone by perception/logic/commonsense — never applicable to fine-grained entity facts (species data, dates, measurements). The fourth verdict maps to *no gating* (outcome-only), which repairs the systematic penalty that entailment-gating inflicts on reasoning-type multiple-choice answers (Appendix G, Case 3), while leaving the anti-guess gate fully active on knowledge-seeking questions. Raw verdicts ship with the dataset for auditability. An offline lexical fallback (answer content-word/number coverage in the evidence, with unmatched numbers vetoing) supports API-free smoke runs; unparseable or unverifiable cases return no verdict and fall back to outcome-only rather than fabricating a value. The judge prompt is given verbatim in Appendix F.

D Dataset Schema and Quality Control

Each released sample carries: identifiers (sample/trajectory/query); the state snapshot before the action (actions so far, evidence bundle); the typed action; the observed results; labels g , R_{out} (with the count of trajectories through the node), *supp* with its raw verdict, the fused score; the rationale, its verdict tag, QC status; and generation provenance (policy model and deployment, retriever, MCTS configuration, seeds, pipeline commit). Hard drop rules (the only ways a sample is excluded): trajectory length outside $[2, 8]$; non-answer steps with $|g - R_{\text{out}}| > 0.7$ (label self-contradiction); rationale QC failure after three regenerations. Failed trajectories are retained — their low-scored steps are the PRM’s negative curriculum — and correct-but-unsupported answers are retained with flags as DPO rejected candidates. Validation enforces score-fusion identities, per-field ranges, the answer $\Leftrightarrow$ null-grounding rule, query-disjoint splits, and redistributable image references. For InfoSeek, answer matching follows the official two-type protocol: string questions use relaxed exact match over the gold set; VALUE questions extract numbers from the prediction and accept any inside the gold range — a deterministic rule, not an LLM judge.

E Construction Pilot Statistics

A 100-query-per-benchmark construction pilot ($P=10$) produced 207 / 170 / 166 deduplicated trajectories and 464 / 366 / 355 step samples on InfoSeek / ScienceQA / GQA respectively, of which 426 / 328 / 341 survived QC (drops dominated by the grounding–outcome gap rule). The pilot surfaced — and motivated the fixes for — three construction-level failure modes: (i) *parametric lucky guesses* on InfoSeek (answers judged correct with zero evidence support), addressed by the support gate of Eq. (6) and illustrated in Appendix G, Case 2; (ii) *systematic entailment penalties* on ScienceQA reasoning questions (105

of the 170 answer steps flagged correct-but-unsupported), addressed by the NOT_REQUIRED verdict (Case 3); and (iii) *action imbalance and trajectory collapse* on InfoSeek (of its 426 retained samples, 206 are answer steps and the 220 retained search steps split 166:54 image:text — inverted against a text-only corpus; and only ≈ 2 deduplicated trajectories survive per query), addressed by retrieval-chain and proposal-diversity constraints in the policy prompt (Appendix F). We report these numbers for transparency: dataset papers rarely publish their construction pathologies, yet they determine label quality.

F Prompts

This appendix reproduces, verbatim, the three prompts used by the system: the policy proposal prompt (inference-time expansion and dataset rollouts), the rationale-generation prompt (dataset Step 3), and the answer-support judge prompt (dataset Step 2). Placeholders in braces are filled at run time.

F.1 Policy Proposal Prompt (Expansion / Rollouts)

```
You are planning retrieval actions to answer a question about the given image.

Question: {question}
Evidence collected so far:
{evidence}

Propose up to {k} candidate next actions as JSON, one per line. Allowed:
{"type": "text_search", "query": "..."} # query the corpus with text
{"type": "image_search"} # query the corpus with the image
{"type": "answer", "text": "..."}

Rules:
- The question refers to the given image (e.g. "this bird", "this building").
  Do NOT invent or assume a specific entity identity (a concrete species,
  landmark, person, ...) that the image and evidence have not established.
  Guessing an identity and searching for it propagates errors through the
  whole trajectory.
- Using an entity name that ALREADY APPEARS IN THE EVIDENCE above is NOT
  guessing: once image_search (or an earlier text_search) has surfaced the
  entity, you SHOULD build text_search queries from that evidence-established
  name.
- Retrieval chain for entity questions: if the entity is not yet identified,
  FIRST use image_search to match the image against the corpus. Once the
  entity is identified but the ASKED ATTRIBUTE is still missing from the
  evidence, the next action MUST include a text_search combining the
  evidence-established entity name with the asked attribute (e.g.
  "<entity name> opening date") -- do NOT answer from an entity match alone.
- Make the candidates genuinely different from each other: text_search
  queries within one proposal must not be near-duplicates -- vary the angle
  (entity name + attribute, bare keywords, a rephrasing of the question), and
  never re-issue a query already executed on this path.
- Keep every search query faithful to the original question's intent; do not
  drift away from what is actually being asked.
- Only answer when the collected evidence actually supports the answer, OR
  when the question is answerable from the image and question alone by
  perception, logic, or everyday commonsense (no external facts needed).
Output ONLY the JSON lines.
```

F.2 Rationale-Generation Prompt (Dataset Step 3)

```
You are writing a training rationale for a process reward model that judges
retrieval actions.

Question: {question}
Evidence collected BEFORE this action:
{evidence_before}
Action taken: {action_type}({action_input})
Evidence retrieved BY this action:
{observation}
Assigned quality score: {score} (0 = poor action, 1 = excellent action)

Write a rationale of 30-150 tokens explaining WHY this score is appropriate.
Requirements:
- explicitly mention the action type "{action_type}";
- cite at least one evidence id in square brackets, e.g. [e0];
- be concrete about how the evidence supports (or fails to support) progress
  toward answering the question;
```

```

- judge the step ONLY from the question, the evidence, and the action -- as if
the final outcome were still unknown. NEVER mention or hint at a gold /
ground-truth answer and never use evaluation language such as "the answer is
correct" or "matches the ground truth";
- if the evidence does not actually support the step's usefulness, say so
plainly.
After the rationale, end with ONE final line of exactly this form (it must be
consistent with the assigned score):
VERDICT: good      (clearly useful step)    or
VERDICT: mixed     (partially useful)      or
VERDICT: poor      (useless or harmful)

```

E.3 Answer-Support Judge Prompt (Dataset Step 2)

```

Question: {question}

Evidence collected so far:
{evidence}

Proposed final answer: {answer}

Classify how this answer relates to the evidence. Reply with exactly one word:

- SUPPORTED: the evidence explicitly contains or entails the answer. For a
multiple-choice answer it is enough that the evidence entails the CONTENT of
the chosen option (verbatim wording is not required). An absence answer like
"no" counts as supported when the evidence describes the relevant
scene/content and it lacks the asked-about item.
- PARTIAL: the evidence points toward the answer but leaves a substantive part
of it unconfirmed.
- UNSUPPORTED: the answer relies on external factual knowledge (entity facts,
numbers, dates, names, statistics) that the evidence does not contain.
- NOT_REQUIRED: the answer is derivable from the question text and the visible
image content alone, by perception, logic, or everyday commonsense -- no
external factual knowledge is needed. Do NOT use this for fine-grained
entity facts (species data, dates, measurements, biographies): those always
require evidence.

```

G Case Studies from the Construction Pilot

The following four cases are drawn from real trajectories of the construction pilot (Appendix E); evidence snippets are abridged. They illustrate, in order: the intended behavior, and the three failure modes that shaped the reward design of Section 3.3.

Case 1 (InfoSeek, success): the disambiguate-then-look-up chain

- **Question:** “When did this building officially open?” (image of the Pro Football Hall of Fame); gold: September 7, 1963.
- **Step 1** TEXT_SEARCH(“opening date Pro Football Hall of Fame”) → passage containing “...opened on September 7, 1963...”. High grounding; on a successful trajectory; kept as a high-score positive.
- **Step 2** IMAGE_SEARCH(query image) → further pages on the same entity (award lists, inductees). Topically related but adds no new information for the asked attribute — see Case 4.
- **Step 3** ANSWER(“September 7, 1963”) → correct *and* supported: the date is present in cited evidence, so $\text{supp}=1$ and the answer step receives a full-score label.

Case 2 (InfoSeek, failure): the parametric lucky guess

- **Question:** “What is the maximum weight this bird usually grows to?” (image of a mourning dove); gold: ≈ 153 g with tolerance.
- **Step 1** TEXT_SEARCH(“how heavy does mourning dove grow up to”) → passages comparing dove species qualitatively; no weight figure. **Step 2** IMAGE_SEARCH → sparrow and dove-call passages; irrelevant. Both search steps have very low grounding on a trajectory whose outcome is 1.0; the $|g - R_{\text{out}}| > 0.7$ rule drops both samples as self-contradictory labels.
- **Step 3** ANSWER(“Up to 145 grams”) → judged *correct* by the tolerance metric, yet *no evidence contains any weight figure*. Without the support gate this step would be labeled 1.0 — a perfect score for answering from parametric memory — and the PRM would learn that evidence-free answering is optimal. With Eq. (6): $\text{supp}=0$ (the unmatched number vetoes), the label

drops to $\gamma \cdot 1.0 = 0.3$, and the sample is flagged `unsupported_correct` for use as a DPO rejected side against sibling actions that kept searching.

Case 3 (ScienceQA, failure): entailment-gating a reasoning answer

- **Question:** “Which question can Jen’s experiment best answer?” (options about radish size vs. leaf count under different light); gold: the size option.
- **Step 1** TEXT_SEARCH(“radishes bright light dim light growth”) → a passage describing the experiment comparing radish *sizes* under two light conditions — decisive for choosing the size option.
- **Step 2** ANSWER(size option) → correct. A pure entailment judge, however, rules UNSUPPORTED: no document states “therefore choose option A” — the answer is *derived* from the evidence and the option semantics, not contained in any span. In the pilot, 105 of the 170 ScienceQA answer steps were flagged correct-but-unsupported this way, pinning them at the 0.3 floor and teaching the PRM a systematic bias against reasoning-type answers. The NOT_REQUIRED verdict (Appendix C) repairs exactly this: the judge classifies the question as answerable from the question, options, and evidence-backed premises by reasoning, and the answer step falls back to its outcome label.

Case 4 (open issue): redundant retrieval on solved states

- In Case 1, Step 2 retrieves same-entity content *after* the asked attribute is already in the bundle. Its grounding is not low — same-entity results are topically relevant — and it lies on a successful trajectory, so it receives a high label from Eq. (5) despite adding nothing: relevance and *marginal utility* diverge. This over-search pattern (cf. Wu et al. 35) is the one construction pathology we deliberately did not patch yet: a naive novelty discount $g' = g \cdot (1 - \max \cos(\text{new evidence}, \mathcal{E}))$ would also punish legitimate multi-source corroboration, which the poisoning robustness of Section 5.3 depends on. We record a marginal-gain field per step in ETBench-Open and defer the labeling decision until the full-scale distributions separate redundancy from corroboration; same-state DPO pairs (answer-now vs. search-more, both successful) provide a label-free training route for *when to stop* in the meantime.

H Provenance of Quoted Baseline Numbers

In Table 1, the published InfoSeek rows use BLIP-2 from Li et al. 19, InstructBLIP from Dai et al. 10, Wiki-LLaVA from Caffagni et al. 6, RoRA-VLM from Qi et al. 27, EchoSight from Yan and Xie 37, ReflectiVA from Cocchi et al. 9, Wiki-PRF from Hong et al. 12, and Wiki-R1 from Ning et al. 26 (unseen-entity split not reported). The ScienceQA block uses Human from Lu et al. 24, MM-CoT from Zhang et al. 44, the fine-tuned LLaVA result from Liu et al. 22, and all Qwen2-VL/InternVL2 zero-shot, Vanilla-RAG, and RCTS rows from Yang et al. 38 under its full-test protocol. The GQA block uses InstructBLIP from Dai et al. 10, u-LLaVA-1.5 from Xu et al. 36, LLaVA-1.5 from Liu et al. 23, BLIP-2 from Li et al. 19, and Qwen-VL-Chat/Qwen-VL from Bai et al. 4. Uncited No-retrieval, Qwen2.5-VL Vanilla-RAG, matched-budget Best-of- N , and EvidenceTree rows are evaluated under our common corpus and inference protocol.

I Notation

Symbol	Meaning
q, I	question; optional query image
$s_t, \mathcal{E}_t$	state; evidence bundle (Eq. 1)
$\mathcal{A}$	typed action space (Eq. 2)
$\tau, y(\tau), y^*$	trajectory; its answer; gold answer
$g(a, s)$	local grounding (Eq. 3)
$R_{\text{out}}(s, a)$	tree-level outcome credit (Eq. 4)
$\text{supp}(y, \mathcal{E})$	answer-support verdict (Eq. 6)
α, γ	fusion weight (0.5); support floor (0.3)
$Q_\phi(a s)$	PRM prior value for a candidate action
$Q(a s)$	running mean of backed-up rewards (Eq. 7), seeded by Q_ϕ
$N(s), N(s, a)$	node / edge visit counts
c	UCB1 exploration constant (1.0)
P, d_{max}, k	rollouts (10); max depth (3); children (3)
η_{stop}	early-stop threshold

Table A2: Notation.